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1. What is a labeled training set?

A labeled training set is a collection of data examples that include both features and the correct output for each example (the label). The example in Hands-On machine learning is a training set of emails labeled with spam, or not spam.

2. What type of algorithm would you use to allow a robot to walk in various unknown terrains?

Reinforcement Learning – and agent (process) can observe the environment, and perform actions, and get rewards or penalties. The agent must learn what is the best strategy to maximize the rewards or minimize the penalties.

3. What type of algorithm would you use to segment your customers into multiple groups?

This is unsupervised learning where we want to discover patterns in data without predefined labels. K-means Clustering is such an algorithm.

4. Would you frame the problem of spam detection as a supervised learning problem or an unsupervised learning problem?

Supervised learning. The algorithm learns from labeled examples in order to classify unseen examples.

5. What is out-of-core learning?

Out-of-core learning is an online learning algorithm to train models on huge datasets. These datasets are too large to fit in one machine's memory. The algorithm loads part of the data, runs a training step on the data, and repeats until all the data has been seen.

6. What is the difference between a model parameter and a model hyperparameter?

Model parameters are estimated in the training process, whereas hyperparameters are set as configurations before training begins. Regularization is discussed in the text as a hyperparameter – a method to constrain model complexity to reduce the risk of overfitting.

7. What do model-based algorithms search for? What is the most common strategy they use to succeed? How do they make predictions?

Model-based algorithms search for optimal parameter values. The most common strategy is to minimize the cost function. Model-based algorithms make predictions by using the trained model parameters on new input data. Prediction =  $w_0 + w_1x_1 + w_2x_2 + \dots$  where the  $w_0$  etc., are the model parameters or weights, and the  $x$  are the new input observed values.

8. If your model performs great on the training data but generalizes poorly to new instances, what is happening? Can you name three possible solutions?

A likely cause is model overfitting. Solutions: 1) get more data, 2) add regularization Hyperparameters to the model configuration, or otherwise reduce model complexity, 3) clean up the existing data – fix errors and remove outliers.

## 9. What is the purpose of a validation set?

The validation set, or holdout set is a subset of the data used to evaluate candidate models selection, and hyperparameter tuning. You train on a test set and select the model that performs best on the validation set.

## 10. What is the train-dev set, when do you need it, and how do you use it?

The train-dev set is a subset of training data used when there is a mismatch between the training data and the data your model will encounter in production. (it's an Andrew Ng thing). You train your model on the training data the evaluate it on the train-dev set. If you identify the model to be overfit, you might get more data, or simplify the model, etc. This was the example where we have web scraped pictures in the training set, but the app model will need to identify pictures from the phone.

## 11. What can go wrong if you tune hyperparameters using the test set?

Your test accuracy will potentially be overstated, and your model will fail on new data.

Your statistical conclusion will be invalid.

## 12. Explain whether each scenario is a classification or regression problem, and indicate whether we are most interested in inference or prediction. Finally, provide n (number of sample points, observations) and p (number of features).

We collect a set of data on the top 500 firms in the US. For each firm we record profit, number of employees, industry and the CEO salary. We are interested in understanding which factors affect CEO salary.

This is Regression. We want to predict CEO salary based on features.  $N = 500$  observations,  $p = 3$  features

We are considering launching a new product and wish to know whether it will be a success or a failure. We collect data on 20 similar products that were previously launched. For each product we have recorded whether it was a success or failure, price charged for the product, marketing budget, competition price, and ten other variables.

This is classification. We interested in predicting success or failure – discrete categorical outcomes. We have 20 observations (similar products) with 13 features (price charged, budget, comp price, + 10 others). Success or failure is the target.

We are interested in predicting the % change in the USD/Euro exchange rate in relation to the weekly changes in the world stock markets. Hence we collect weekly data for all of 2012. For each week we record the % change in the USD/Euro, the change in the US market, the % change in the British market, and the % change in the German market.

This is regression, we are interested in predicting the % change in FX rate given change in stock markets. We have 52 weekly observations of USD/EUR and US Stocks, UK Stocks, and GDR Stocks, so  $p = 3$  features.

13. Read the paper Scaling to Very Very Large Corpora for Natural Language Disambiguation by Banko and Brill and write a brief summary.

More data is better. The paper looks to test if increasing the size of the training data affects the performance in specific tasks. Turns out it does. An important finding is that more data beats smarter algorithms. Voting works best on a smaller scale and quickly loses impact. Active learning may provide a path to reducing annotation cost (more human labeled data). They propose some schemes to automate labeling.

14. The table below provides a training data set containing six observations, three predictors, and one qualitative response variable.

Suppose we wish to use this data set to make a prediction for Y when  $X_1 = X_2 = X_3 = 0$  using K-nearest neighbors.

Compute the Euclidean distance between each observation and the test point,  $X_1 = X_2 = X_3 = 0$ .

```
distances = np.sqrt(np.sum((feature_data - test_point)**2, axis=1))
```

Observations sorted by distance from test point (0,0,0):

| Index | X1 | X2 | X3 | Y     | distance |
|-------|----|----|----|-------|----------|
| 4     | -1 | 0  | 1  | green | 1.414214 |
| 5     | 1  | 1  | 1  | red   | 1.732051 |
| 1     | 2  | 0  | 0  | red   | 2.000000 |
| 3     | 0  | 1  | 2  | green | 2.236068 |
| 0     | 0  | 3  | 0  | red   | 3.000000 |
| 2     | 0  | 1  | 3  | red   | 3.162278 |

What is our prediction with K = 1? Why?

K=1 Prediction: green

Reason: The closest observation is at distance 1.4142

What is our prediction with K = 3? Why

K=3 Prediction: red

Reason: Average of 3 nearest neighbors:

Neighbor 1: (-1, 0, 1) -> Y=green, distance=1.4142

Neighbor 2: (1, 1, 1) -> Y=red, distance=1.7321

Neighbor 3: (2, 0, 0) -> Y=red, distance=2.0000